
Proposed Project Title

AI-Powered Smart Helmet for Predictive Rider Safety and Emergency Response

This title emphasizes that your helmet not only reacts to accidents but also **predicts and prevents them**.

Feature Set

I'll divide the features into **three levels**.

Level 1 – Basic IoT Features (Must Have)

These are essential because almost every smart helmet includes them.

Feature	IoT	AI	Include?
Helmet Detection			
Accident Detection			
GPS Tracking			
SOS Alert (SMS/App)			

Bike Ignition Control   

Mobile App   

Cloud Dashboard   

Level 2 – AI Features (Your Uniqueness)

These are what make your project different.

1. AI Drowsiness Detection

Uses

- ESP32-CAM / Camera
- MediaPipe
- TensorFlow Lite

Detects:

- Eye closure
- Yawning
- Head nodding

If sleepy:

- Helmet vibrates
 - Voice alert
 - Dashboard updated
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2. AI Distraction Detection

Detect:

- Looking at phone
- Looking away from road
- Eyes closed too long

Warn the rider before an accident occurs.

3. AI Safety Score

After every ride:

Helmet Used ✓

Over Speed 1

Hard Brake 2

Sharp Turn 0

Fatigue Alert 1

Safety Score

91 /100

Parents and companies can monitor rider behavior.

4. AI Risk Prediction ★★★★★

Instead of waiting for an accident,

AI predicts dangerous riding.

Inputs:

- Speed
- Lean angle
- Vibration
- Fatigue
- Hard braking

Output:

High Risk Riding

Warning!

This is predictive safety.

5. AI Voice Assistant ★★★★★☆

Commands:

Call Home

Send SOS

Navigate to College

Battery Status

Weather

Nearest Petrol Pump

Level 3 – Advanced IoT Features

Rider Health Monitoring

Sensors:

- MAX30102

Detect:

- Heart Rate
- SpO₂

If abnormal:

Emergency Alert

↓

GPS

↓

Hospital



Parents

Environmental Monitoring ★★★★★☆

Sensors:

- MQ135

Detect:

- Air Pollution
- Toxic Gas

Helmet says

"Air quality is poor."

Weather Detection ★★★★★☆

Use weather API.

Helmet says

"Heavy rain ahead."

Road Hazard Detection ★★★★★

Using camera

AI detects:

- Potholes
- Speed breakers
- Fallen trees
- Waterlogging

Helmet warns rider.

Hazard Sharing ★★★★★

Suppose Rider A detects

Pothole

↓

Upload GPS

↓

Cloud

↓

All riders receive

Warning

Pothole 200m Ahead

Live Dashboard ★★★★★

Dashboard shows

Live Location

Battery

Helmet Status

Heart Rate

Speed

Safety Score

Accident Alerts

Ride History

Black Box ★★★★★

Store last 60 seconds

GPS

Speed

Heart Rate

Video

Lean Angle

Useful after accidents.

AI Models

AI Feature	Model
Helmet Detection	YOLOv11
Drowsiness	MediaPipe
Face Detection	OpenCV
Fatigue	TensorFlow Lite
Safety Prediction	Random Forest / XGBoost
Voice Assistant	Whisper / Vosk

IoT Hardware

Component	Purpose
ESP32	Main Controller
ESP32-CAM	AI Camera
MPU6050	Tilt & Accident Detection
MAX30102	Heart Rate & SpO ₂
MQ3	Alcohol Detection
GPS NEO-6M	Live Location
SIM800L	GSM Alert
Buzzer	Voice Alert
Vibration Motor	Haptic Warning
Li-ion Battery	Power

Software Stack

AI

- Python
- TensorFlow Lite
- OpenCV
- MediaPipe
- YOLOv11

IoT

- Arduino IDE
- ESP32
- MQTT

Backend

- Node.js
- Express.js

Database

- MongoDB

Frontend

- React.js

Mobile App

- Flutter *(or React Native if you prefer to stay in JavaScript)*

Cloud

- Firebase
 - MQTT Broker
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Architecture

Helmet



ESP32



Sensors



AI Processing



Wi-Fi / GSM



Cloud



Dashboard



Parents / Hospital

Innovation Score

Feature	Existing Papers	Your Project
Helmet Detection	✓	✓
Accident Detection	✓	✓
GPS Tracking	✓	✓
AI Drowsiness	✗	✓
AI Risk Prediction	✗	✓
Safety Score	✗	✓
Rider Health	✗	✓
Hazard Sharing	✗	✓

Black Box	✗	✓
Voice Assistant	Few	✓
Dashboard Analytics	Basic	Advanced

My recommended feature set

For a **final-year project** that is achievable and demonstrates both **IoT** and **AI**, I recommend these **10 core features**:

1. Helmet Detection
2. Alcohol Detection
3. Accident Detection
4. GPS + GSM Emergency Alert
5. **AI Drowsiness Detection**
6. **Rider Health Monitoring (Heart Rate + SpO₂)**
7. **AI Safety Score**
8. Live IoT Dashboard
9. Voice Assistant for hands-free commands
10. Black Box Ride Recorder

This combination keeps the hardware manageable while clearly showcasing both embedded IoT engineering and practical AI. It also differentiates your project from the research papers you reviewed by emphasizing **accident prevention**, **rider behavior analysis**, and **health-aware safety**, rather than only responding after a crash.